

Artificial Intelligence Application Development Practical

AI-Powered Heart Disease Risk Analysis and Prediction

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1. INTRODUCTION

The AI-Powered Heart Disease Risk Analysis and Prediction project is an AI-based application that uses Machine Learning to predict the estimated risk of heart disease. It analyzes health-related factors such as age, blood pressure, cholesterol level, BMI, smoking habits, and physical activity. The system uses the Random Forest Classification algorithm to generate a risk prediction.

The project addresses the difficulty of analyzing multiple health-related factors together for early risk assessment. It provides an easy-to-use system where health information can be entered and analyzed to obtain an estimated heart disease risk level and probability.

The project is useful for demonstrating how Artificial Intelligence and Machine Learning can be applied to healthcare risk assessment. It also displays important contributing factors and model performance measures such as Accuracy, Precision, Recall, and F1 Score, helping users understand the application of machine learning in a healthcare context.

2. PROBLEM STATEMENT

The Heart disease is a major health concern, and identifying its risk factors at an early stage can be helpful.

Traditional assessment may require analyzing multiple health-related factors such as age, blood pressure, cholesterol, BMI, smoking habits, and physical activity.

Therefore, the problem is to develop an easy-to-use AI-based system that can analyze these health-related factors and provide an estimated heart disease risk level and probability. The project aims to demonstrate how Artificial Intelligence and Machine Learning can support early risk assessment and help users understand the factors associated with the prediction.

3. OBJECTIVES

The main objectives of the AI-Powered Heart Disease Risk Analysis and Prediction project are:

1. To collect and prepare health-related data required for heart disease risk prediction.
2. To preprocess the data and prepare it for machine learning.
3. To apply the Random Forest Classification algorithm to the prepared data.
4. To train and test the machine learning model using the dataset.
5. To generate heart disease risk predictions along with the estimated risk probability.

4. TECHNOLOGIES USED

The following technologies and software are used in the project:

- Python – Programming language
- Pandas – Data handling and processing
- Scikit-learn – Machine learning model development and evaluation
- Random Forest – Classification algorithm
- Streamlit – Web-based user interface
- Visual Studio Code (VS Code) – Development environment

5. DATASET

The project uses a health-related dataset containing information required to predict the estimated risk of heart disease.

- Dataset Name: Heart Disease Risk Prediction Dataset
- Source of Dataset: The project document does not specify an external dataset source. The project uses health-related data for training the Random Forest model.
- Number of Records: 20 records were used in the project's sample training dataset.

- Important Attributes/Features:
 - Age
 - Blood Pressure
 - Cholesterol
 - BMI
 - Smoking
 - Physical Activity
- Target Variable: Risk — indicates the heart disease risk category used for classification.

These features are used as inputs to train the Random Forest Classification model, which then generates the estimated heart disease risk and probability.

6. METHODOLOGY/WORKING

The following steps were followed to develop the **AI-Powered Heart Disease Risk Analysis and Prediction** system:

1. **Data Collection**
Health-related sample data was created with factors such as Age, Blood Pressure, Cholesterol, BMI, Smoking, and Physical Activity.
2. **Data Preparation**
The data was stored in a Pandas DataFrame. The input features were separated from the target variable Risk.
3. **Data Splitting**
The dataset was divided into 75% training data and 25% testing data using `train_test_split()`.
4. **Model Training**
A Random Forest Classification algorithm was used to train the model on the health-related input features.
5. **Risk Prediction**
The trained model accepts patient information and predicts whether the estimated risk is lower or higher.
6. **Risk Probability and Level**
The system calculates the probability of higher risk and classifies it as LOW, MODERATE, or HIGH based on predefined probability ranges.
7. **Feature Importance Analysis**
Random Forest feature importance is used to display which input factors contribute most to the model's prediction.
8. **Model Evaluation**
The model is evaluated using Accuracy, Precision, Recall, and F1 Score.
9. **Streamlit Interface**
A user-friendly interface was developed using Streamlit, allowing users to enter health information and view the prediction, probability, risk level, and model performance.
10. **Result and Disclaimer**
The system displays the prediction and analysis results. It is intended for educational purposes only and not as a medical diagnosis.

7. ALGORITHM USED

Random Forest Classification

Random Forest is a supervised machine learning classification algorithm that combines multiple decision trees to make a prediction. Each tree gives a prediction, and the final result is decided based on the majority of the tree predictions.

Random Forest is used to predict the heart disease risk based on multiple health-related factors such as Age, Blood

Pressure, Cholesterol, BMI, Smoking, and Physical Activity. It is suitable for classification problems where the output belongs to categories such as lower or higher risk.

1. The available health data is divided into training and testing data.
2. Multiple decision trees are created using different portions of the training data.
3. Each tree analyzes the patient's input features and makes a prediction.
4. The predictions from all trees are combined.
5. The Random Forest produces the final risk prediction and probability.
6. The model's performance is evaluated using Accuracy, Precision, Recall, and F1 Score.

In this project, the Random Forest model is created using RandomForestClassifier with 100 decision trees.

8. IMPLEMENTATION

Exploratory Data Analysis (EDA) is performed to understand the structure, characteristics and patterns present in the Retail and warehouse Sale.csv dataset. The dataset is explored using Python and suitable visualizations.

CODE:

```
import streamlit as st
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
# PAGE CONFIGURATION
st.set_page_config(
    page_title="AI Heart Risk Predictor",
    page_icon="❤️",
    layout="wide"
)
# TITLE
st.title("❤️ AI-Based Heart Disease Risk Prediction")
st.write(
    "An educational AI system that predicts heart disease "
    "risk using a Random Forest classification model."
    "A Group Project Made By Raj,Khushi,Priti"
)
# SAMPLE DATASET
data = {
    "Age": [
        25, 32, 45, 52, 61,
        29, 40, 55, 67, 36,
        48, 58, 23, 70, 43,
        51, 63, 31, 46, 57
    ],
    "BloodPressure": [
        110, 120, 140, 150, 160,
        115, 130, 145, 170, 125,
        135, 155, 105, 180, 128,
        148, 165, 118, 138, 152
    ],
    "Cholesterol": [
        170, 180, 230, 250, 270,
```

```

        160, 210, 240, 290, 190,
        220, 260, 155, 300, 205,
        245, 280, 175, 225, 255
    ],
    "BMI": [
        21, 23, 28, 30, 32,
        20, 25, 29, 34, 24,
        27, 31, 19, 35, 26,
        30, 33, 22, 28, 32
    ],
    "Smoking": [
        0, 0, 1, 1, 1,
        0, 0, 1, 1, 0,
        1, 1, 0, 1, 0,
        1, 1, 0, 0, 1
    ],
    "PhysicalActivity": [
        1, 1, 0, 0, 0,
        1, 1, 0, 0, 1,
        0, 0, 1, 0, 1,
        0, 0, 1, 1, 0
    ],
    "Risk": [
        0, 0, 1, 1, 1,
        0, 0, 1, 1, 0,
        1, 1, 0, 1, 0,
        1, 1, 0, 0, 1
    ]
}]
df = pd.DataFrame(data)
# PREPARE DATA
X = df.drop("Risk", axis=1)
y = df["Risk"]
# TRAIN AND TEST DATA
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.25,
    random_state=42,
    stratify=y)
# CREATE RANDOM FOREST MODEL
model = RandomForestClassifier(
    n_estimators=100,
    random_state=42)
# TRAIN MODEL
model.fit(X_train, y_train)
# MODEL PREDICTION
y_pred = model.predict(X_test)
# MODEL PERFORMANCE
accuracy = accuracy_score(y_test, y_pred)

```

```

precision = precision_score(
    y_test,
    y_pred,
    zero_division=0)
recall = recall_score(
    y_test,
    y_pred,
    zero_division=0
)
f1 = f1_score(
    y_test,
    y_pred,
    zero_division=0
)
# PATIENT INFORMATION
st.header("👤 Enter Patient Information")
col1, col2 = st.columns(2)
# LEFT COLUMN
with col1:
    age = st.number_input(
        "Age",
        min_value=18,
        max_value=100,
        value=40 )
    bp = st.number_input(
        "Blood Pressure",
        min_value=80,
        max_value=220,
        value=120
    )
    cholesterol = st.number_input(
        "Cholesterol",
        min_value=100,
        max_value=400,
        value=200
    )
# RIGHT COLUMN
with col2:
    bmi = st.number_input(
        "BMI",
        min_value=10.0,
        max_value=50.0,
        value=24.0
    )
    smoking = st.selectbox(
        "Smoking",
        ["No", "Yes"]
    )
    activity = st.selectbox(

```

```

        "Regular Physical Activity",
        ["Yes", "No"]
    )
# CONVERT INPUT VALUES
smoking_value = 1 if smoking == "Yes" else 0
activity_value = 1 if activity == "Yes" else 0
# PREDICTION BUTTON
if st.button("🔍 Predict Heart Disease Risk"):
    input_data = pd.DataFrame([
        {
            "Age": age,
            "BloodPressure": bp,
            "Cholesterol": cholesterol,
            "BMI": bmi,
            "Smoking": smoking_value,
            "PhysicalActivity": activity_value
        } ])
    # MAKE PREDICTION
    prediction = model.predict(input_data)[0]
    probability = model.predict_proba(input_data)[0][1]
    # PREDICTION RESULT
    st.subheader("📄 Prediction Result")
    if prediction == 1:
        st.error(
            "⚠️ Higher Heart Disease Risk Detected"
        )
    else:
        st.success(
            "✅ Lower Heart Disease Risk Detected" )
    # RISK PROBABILITY
    st.write(
        f"Estimated Risk Probability: "
        f"**{probability * 100:.2f}%**"
    )
    # RISK LEVEL
    if probability < 0.35:
        risk_level = "LOW"
    elif probability < 0.65:
        risk_level = "MODERATE"
    else:
        risk_level = "HIGH"
    st.info(
        f"Risk Level: **{risk_level}**"
    )
    # PROGRESS BAR
    st.progress(
        float(probability)
    )
    # EXPLAINABLE AI

```

```

st.subheader("🧠 Important Factors")
importance = pd.DataFrame(
    {
        "Factor": X.columns,
        "Importance": model.feature_importances_
    })
importance = importance.sort_values(
    "Importance",
    ascending=False )

st.bar_chart(
    importance.set_index("Factor") )
# SHOW FACTOR TABLE
st.write("Feature Importance:")
st.dataframe(
    importance,
    use_container_width=True )
# MODEL PERFORMANCE
st.header("📊 AI Model Performance")
c1, c2, c3, c4 = st.columns(4)
with c1:
    st.metric(
        "Accuracy",
        f"{accuracy * 100:.2f}%")
with c2:
    st.metric(
        "Precision",
        f"{precision * 100:.2f}%" )
with c3:
    st.metric(
        "Recall",
        f"{recall * 100:.2f}%")
with c4:
    st.metric(
        "F1 Score",
        f"{f1 * 100:.2f}%"
    )
# DATASET PREVIEW
st.header("📁 Training Dataset")
with st.expander("View Training Dataset"):
    st.dataframe(
        df,
        use_container_width=True
    )
# PROJECT INFORMATION
st.header("📖 About This Project")
st.write("""This project uses a Random Forest Classification algorithm
to predict heart disease risk based on age, blood pressure,
cholesterol, BMI, smoking habits and physical activity.

```

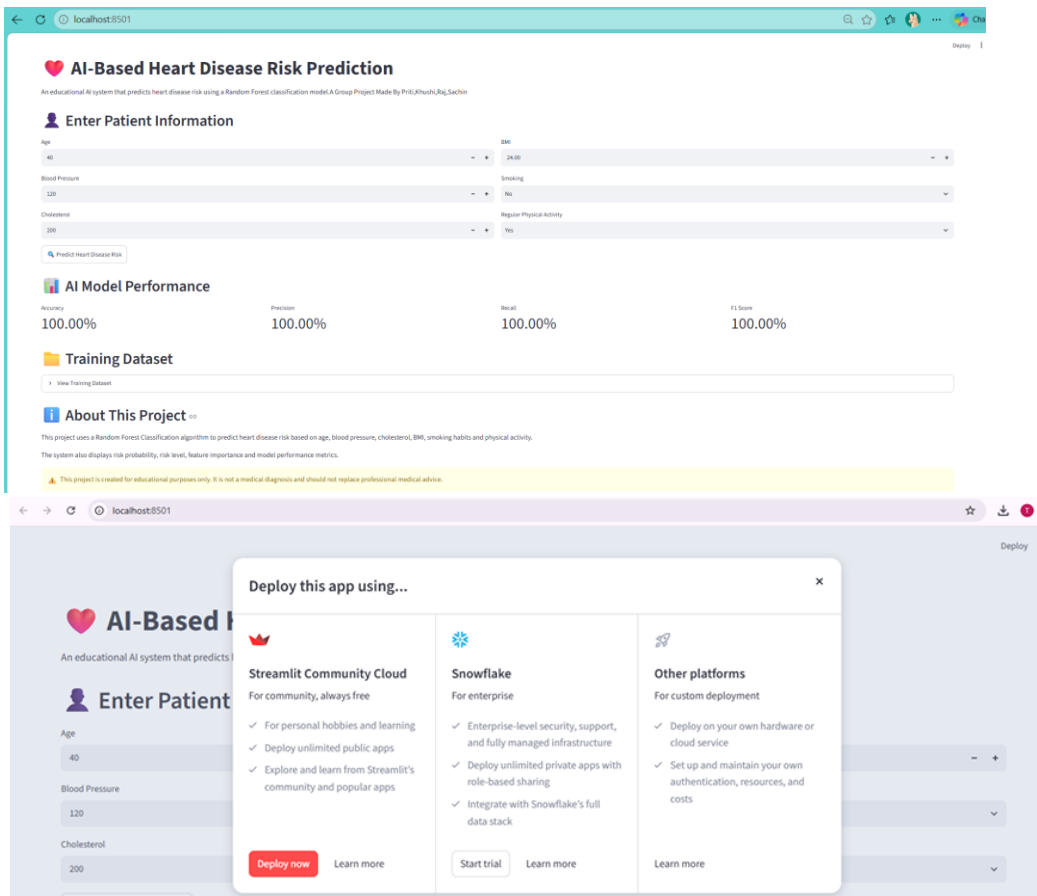
The system also displays risk probability, risk level, feature importance and model performance metrics.

```
""")
# =====
# DISCLAIMER
# =====
st.warning(
    "⚠️ This project is created for educational purposes only. "
    "It is not a medical diagnosis and should not replace "
    "professional medical advice.")
```

CODE EXPLANATION:

The project was implemented using Python, Pandas, Scikit-learn, Random Forest, and Streamlit. First, the health-related sample data containing Age, Blood Pressure, Cholesterol, BMI, Smoking, Physical Activity, and Risk was stored in a Pandas DataFrame. The input features were separated from the target variable Risk. The dataset was then divided into 75% training data and 25% testing data. A Random Forest Classifier with 100 decision trees was trained using the training data. After training, the user enters their health information through the Streamlit interface. The entered values are converted into the required format and provided to the trained model. The model then predicts whether the estimated heart disease risk is lower or higher and calculates the risk probability. Based on this probability, the system displays a LOW, MODERATE, or HIGH risk level. The application also displays the important contributing factors using feature importance. Finally, the model is evaluated using Accuracy, Precision, Recall, and F1 Score. Streamlit provides the user-friendly interface through which all inputs, predictions, charts, and results are displayed. The implementation is intended for educational purposes and not for medical diagnosis.

OUTPUT:



← → ↻ signup.snowflake.com/?trial=streamlit&cloudp=aws&utm_source=streamlit&utm_medium=referral&utm_campaign=na-us-en-5&utm_content=-ss-streamlit-local-deploy ☆ ⬇️ 🔔

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 **Check your inbox!**

An email to activate your account has been sent to pritiyadavmlu@gmail.com. It may take a few minutes to arrive.

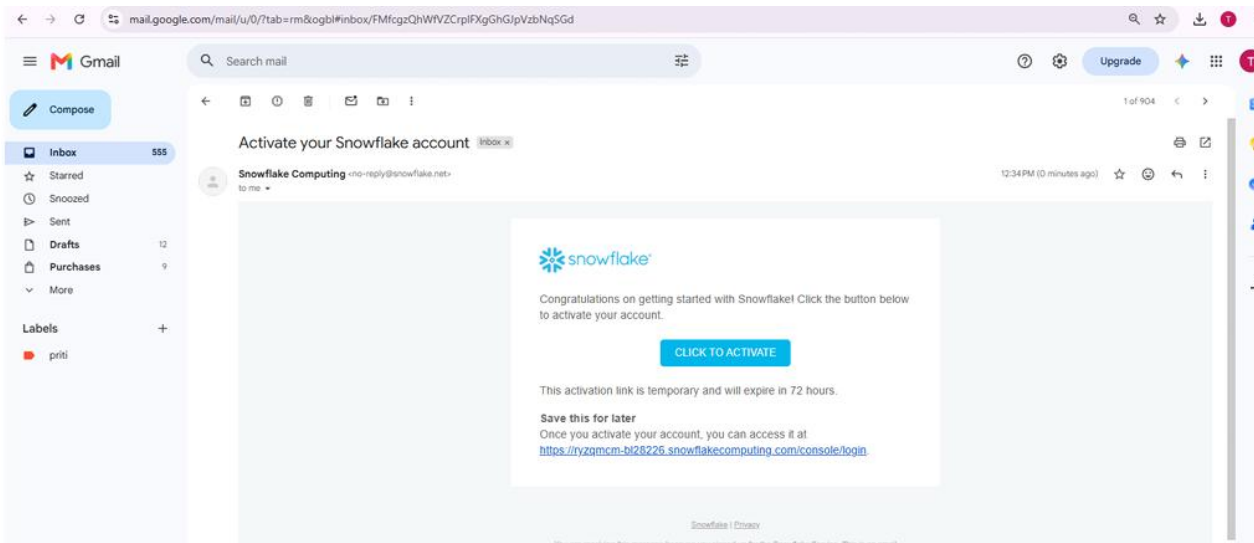
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```

PS C:\Users\priti yadav\OneDrive\Desktop\AI Prac> pip install streamlit pandas scikit-learn
Collecting streamlit
  Downloading streamlit-1.65.0-py3-none-any.whl.metadata (10 kB)
Requirement already satisfied: pandas in C:\Users\priti yadav\AppData\Local\Programs\Python\Python312\Lib\site-packages (3.0.1)
Requirement already satisfied: scikit-learn in C:\Users\priti yadav\AppData\Local\Programs\Python\Python312\Lib\site-packages (1.9.0)
Collecting altair!=5.4.0,!=5.4.1,<7,>=5.0.0 (from streamlit)
  Downloading altair-6.3.0-py3-none-any.whl.metadata (11 kB)
PS C:\Users\priti yadav\OneDrive\Desktop\AI Prac> python -m pip install streamlit pandas scikit-learn
Requirement already satisfied: streamlit in C:\Users\priti yadav\AppData\Local\Programs\Python\Python312\Lib\site-packages (1.65.0)
Requirement already satisfied: pandas in C:\Users\priti yadav\AppData\Local\Programs\Python\Python312\Lib\site-packages (3.0.1)
Requirement already satisfied: scikit-learn in C:\Users\priti yadav\AppData\Local\Programs\Python\Python312\Lib\site-packages (1.9.0)
Requirement already satisfied: altair!=5.4.0,!=5.4.1,<7,>=5.0.0 in C:\Users\priti yadav\AppData\Local\Programs\Python\Python312\Lib\site-packages (6.3.0)
Requirement already satisfied: click<9,>=7.0 in C:\Users\priti yadav\AppData\Local\Programs\Python\Python312\Lib\site-packages (8.1.8)
Requirement already satisfied: numpy<3, >=1.23 in C:\Users\priti yadav\AppData\Local\Programs\Python\Python312\Lib\site-packages (2.0.2)
Requirement already satisfied: packaging in C:\Users\priti yadav\AppData\Local\Programs\Python\Python312\Lib\site-packages (24.1)
Requirement already satisfied: pyarrow in C:\Users\priti yadav\AppData\Local\Programs\Python\Python312\Lib\site-packages (16.1.0)
Requirement already satisfied: python-dateutil in C:\Users\priti yadav\AppData\Local\Programs\Python\Python312\Lib\site-packages (2.9.0)
Requirement already satisfied: requests in C:\Users\priti yadav\AppData\Local\Programs\Python\Python312\Lib\site-packages (2.32.0)
Requirement already satisfied: tenacity in C:\Users\priti yadav\AppData\Local\Programs\Python\Python312\Lib\site-packages (9.0.0)
Requirement already satisfied: toml in C:\Users\priti yadav\AppData\Local\Programs\Python\Python312\Lib\site-packages (0.10.2)
Requirement already satisfied: typing-extensions in C:\Users\priti yadav\AppData\Local\Programs\Python\Python312\Lib\site-packages (4.12.2)
Requirement already satisfied: urllib3 in C:\Users\priti yadav\AppData\Local\Programs\Python\Python312\Lib\site-packages (2.2.3)
Requirement already satisfied: watchdog in C:\Users\priti yadav\AppData\Local\Programs\Python\Python312\Lib\site-packages (4.0.2)
Requirement already satisfied: xyzservices in C:\Users\priti yadav\AppData\Local\Programs\Python\Python312\Lib\site-packages (2024.9.0)
PS C:\Users\priti yadav\OneDrive\Desktop\AI Prac> streamlit run app.py
Note: Global skills include a discover.py script that finds
      project-specific bundled skills at runtime.
2026-10-03 22:39:48.073 Please replace `use_container_width` with `width`.

`use_container_width` will be removed after 2025-12-31.

For `use_container_width=True`, use `width='stretch'`. For `use_container_width=False`, use `width='content'`.
Stopping...
PS C:\Users\priti yadav\OneDrive\Desktop\AI Prac> streamlit run app.py
2026-10-03 22:40:49.946 Uvicorn server started on :::8501

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://192.168.1.9:8501

2026-10-03 22:40:54.221 Please replace `use_container_width` with `width`.

`use_container_width` will be removed after 2025-12-31.

For `use container width=True`, use `width='stretch'`. For `use container width=False`, use `width='content'`.

```

RESULT

1. Accuracy / Performance

The Random Forest model was trained using the prepared dataset and evaluated using Accuracy, Precision, Recall, and F1 Score. These metrics were used to measure how well the model performed on the test data.

The results are displayed directly in the Streamlit application under “AI Model Performance.”

Performance metrics included:

- Accuracy
- Precision

- Recall
- F1 Score

2. Prediction Results

The system successfully predicts the estimated heart disease risk based on the user's:

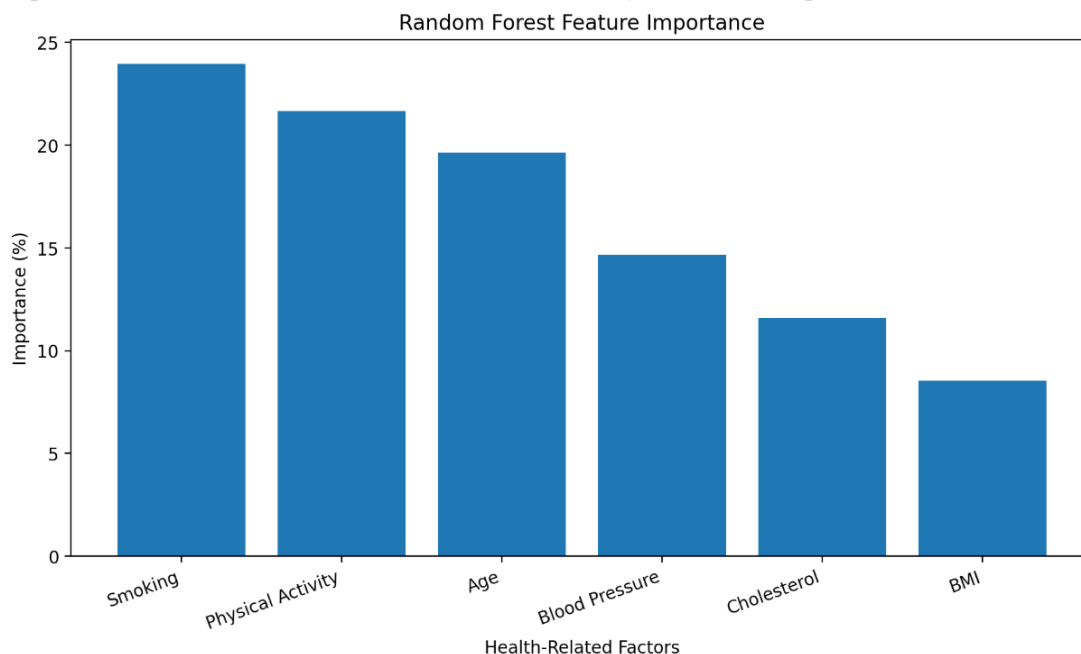
- Age
- Blood Pressure
- Cholesterol
- BMI
- Smoking
- Physical Activity

The output displays:

- Higher / Lower Heart Disease Risk
- Estimated Risk Probability
- Risk Level – LOW / MODERATE / HIGH

3. Graphs

The project generates a Feature Importance Bar Chart using the Random Forest model. The graph shows the relative importance of the different health-related factors used by the model for prediction.



Overall Result

The project successfully implemented an AI-based heart disease risk prediction system that accepts health-related inputs, applies a Random Forest classification model, and displays the estimated risk, probability, important factors, graphs, and model performance through an interactive Streamlit interface.

ADVANTAGES

1. **Easy to Use:** The Streamlit interface allows users to enter health information easily and view the prediction.
2. **Multiple Health Factors:** The system considers Age, Blood Pressure, Cholesterol, BMI, Smoking, and Physical Activity.
3. **Risk Probability:** It provides an estimated probability along with the predicted risk level.

4. **Feature Importance:** The system shows the relative importance of different factors used by the Random Forest model.
5. **Model Evaluation:** Accuracy, Precision, Recall, and F1 Score are displayed to evaluate the model.

LIMITATIONS

1. The project uses a small, manually created sample dataset of 20 records, so the results should not be considered clinically reliable.
2. The system uses only a limited number of health-related factors.
3. The prediction is an estimated machine-learning output and is not a medical diagnosis.
4. The model would require a larger and more representative real-world dataset for practical healthcare use.

APPLICATIONS

The proposed system can be used for:

- Educational demonstrations of AI and Machine Learning in healthcare.
- Student projects and practical applications to understand classification algorithms.
- Health-risk awareness applications for demonstrating how different health factors can be analyzed.
- Prototype healthcare applications where machine-learning prediction concepts need to be demonstrated.
- Future healthcare decision-support research, after proper validation with appropriate clinical data.

CONCLUSION

The project “AI-Powered Heart Disease Risk Analysis and Prediction” successfully developed an interactive machine-learning application using Python, Random Forest, Pandas, Scikit-learn, and Streamlit.

The system accepts health-related parameters such as Age, Blood Pressure, Cholesterol, BMI, Smoking, and Physical Activity and uses a Random Forest classification model to generate an estimated heart disease risk, probability, and risk level. It also provides feature importance and model performance metrics.

AI was used to learn patterns from the prepared sample data and generate a classification prediction based on the user's input. Thus, the project demonstrates the practical application of AI/ML concepts in healthcare risk analysis.

FUTURE SCOPES

The project can be improved in the future by:

1. Using a larger and reliable real-world healthcare dataset.
2. Adding more health parameters such as family history, diabetes, exercise, heart rate, and other relevant clinical factors.
3. Adding user login and secure patient-data storage.
4. Connecting the application to a database for storing patient records.
5. Improving the interface with a more advanced GUI/dashboard.
6. Adding stronger input validation and data preprocessing.
7. Comparing Random Forest with other machine-learning algorithms.
8. Conducting proper validation before considering any real-world healthcare application.

REFERENCES

The following resources were used for understanding and implementing the project:

1. Python Documentation – Python programming language reference.
2. Pandas Documentation – Used for creating and handling the DataFrame and dataset.
3. Scikit-learn Documentation – Used for train_test_split, RandomForestClassifier, and model evaluation.
4. Streamlit Documentation – Used for developing the interactive web application and displaying inputs, outputs, charts, and metrics.
5. GitHub Repository – Project source code and implementation files.
6. Project-created Heart Disease Risk Sample Dataset – The dataset used in this project was manually created within the Python code and is not an externally sourced named dataset.

GITHUB

<https://github.com/pritiyadavmvlu-cloud/AI-Application-Development/blob/main/app.py>